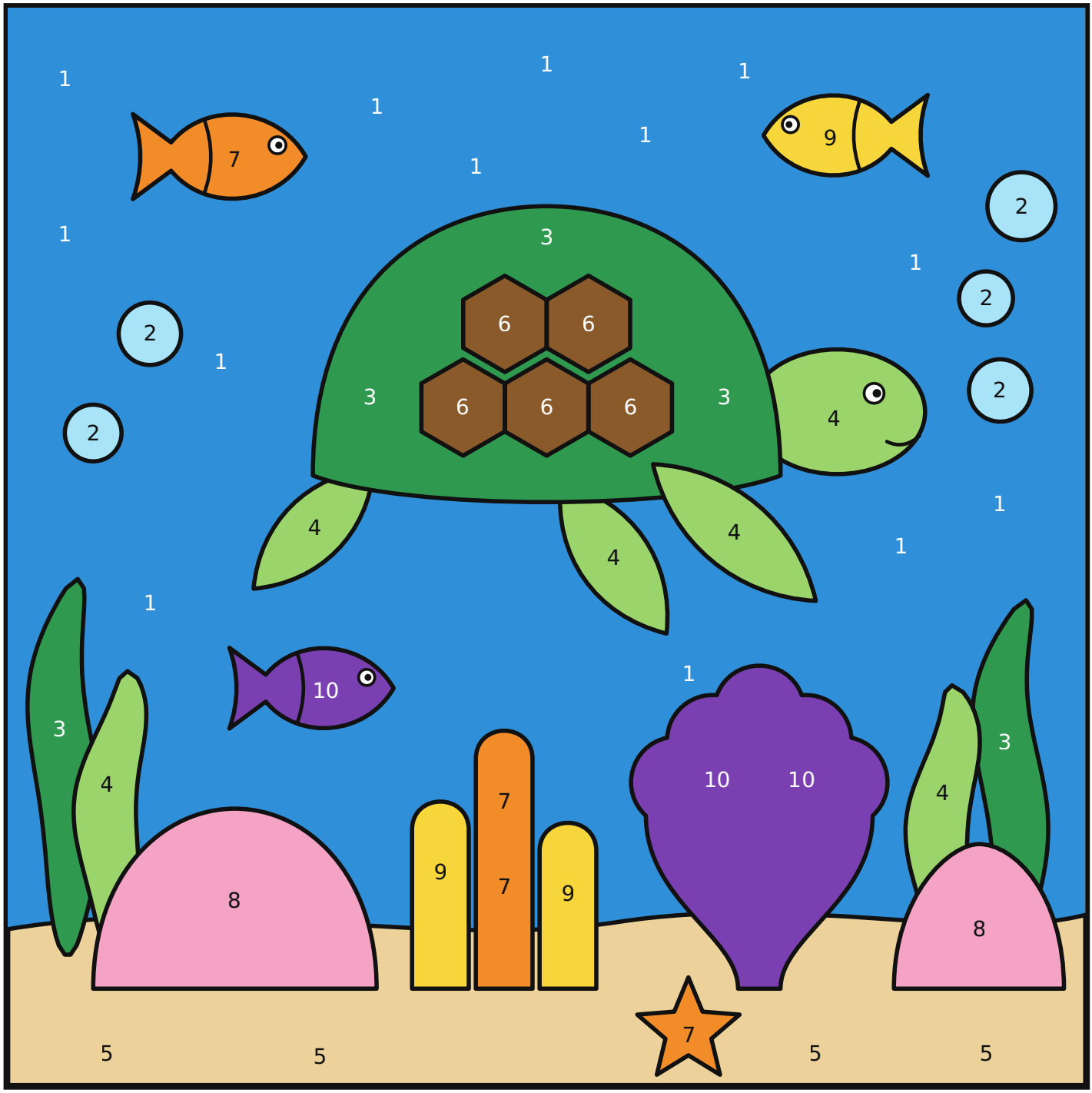
Name: **ANSWER KEY** _____ Date: _____

Graphing Inequalities: Color By Number



Color Code:

8 Pink $y \leq \frac{1}{2}x - 1$	**7** Orange $y > \frac{1}{2}x - 1$	**6** Brown $y \leq -2x - 1$	**5** Tan $y > 2x - 1$	**3** Green $y < -2x + 1$
1 Blue $y > 2x + 1$	**10** Purple $y \geq -\frac{1}{2}x + 2$	**9** Yellow $y < -\frac{1}{2}x + 2$	**2** Light Blue $y \leq 2x + 1$	**4** Light Green $y \geq -2x + 1$

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Graphing Inequalities: Color By Number

Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

1.
A **dashed** boundary line passes through (0, 1) and (1, 3). The region **above** the line is shaded.

slope = 2, y-intercept = 1
dashed → >, above → >
$y > 2x + 1$

1. Color: **Blue**

2.
A **solid** boundary line passes through (0, 1) and (1, 3). The region **below** the line is shaded.

slope = 2, y-intercept = 1
solid → ≤, below → ≤
$y \leq 2x + 1$

2. Color: **Light Blue**

3.
A **dashed** boundary line passes through (0, 1) and (1, −1). The region **below** the line is shaded.

slope = −2, y-intercept = 1
$y < -2x + 1$

3. Color: **Green**

4.
A **solid** boundary line passes through (0, 1) and (−1, 3). The region **above** the line is shaded.

slope = $\frac{3-1}{-1-0} = -2$, y-intercept = 1
$y \geq -2x + 1$

4. Color: **Light Green**

5.
A **dashed** boundary line has slope 2 and y-intercept −1. The region **above** the line is shaded.

dashed, above → >
$y > 2x - 1$

5. Color: **Tan**

6.
A **solid** boundary line has slope −2 and y-intercept −1. The region **below** the line is shaded.

solid, below → ≤
$y \leq -2x - 1$

6. Color: **Brown**

7.
A **dashed** boundary line passes through (0, −1) and (2, 0). The point (0, 0) is in the shaded region.

slope = $\frac{1}{2}$
(0, 0) is above the line (0 > −1), so shade above
$y > \frac{1}{2}x - 1$

7. Color: **Orange**

8.
A **solid** boundary line passes through (0, −1) and (−2, −2). The shaded region is **below** the line.

slope = $\frac{-2-(-1)}{-2-0} = \frac{1}{2}$
$y \leq \frac{1}{2}x - 1$

8. Color: **Pink**

9.
A **dashed** boundary line passes through (0, 2) and (2, 1). The point (0, 0) is in the shaded region.

slope = $-\frac{1}{2}$
(0, 0) is below the line (0 < 2), so shade below
$y < -\frac{1}{2}x + 2$

9. Color: **Yellow**

10.
A **solid** boundary line passes through (0, 2) and (−2, 3). The shaded region is **above** the line.

slope = $\frac{3-2}{-2-0} = -\frac{1}{2}$
$y \geq -\frac{1}{2}x + 2$

10. Color: **Purple**